Mend-Amar Badral

MendeBadra | Mend-Amar Badral | mendebadra.github.io
mendamar.badral@gmail.com | +36 70 7385 744 | Budapest, Hungary

Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop high quality solutions for the real-world, impactful problems.

Education

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: Márton Vaitkus
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0 **(rank 3/34)**
Supervisor: Galtbayar Artbazar
Thesis: *Evaluating land degradation using image processing methods*

Experience

Center of Mathematics Application Research Lab Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- **Geospatial QA & Analysis:** Developed a custom **Python program** using **pandas** and **rasterio** to process drone-captured location datasets. Integrated camera and sensor calibration parameters to ensure data accuracy, utilizing Moran's I coefficient for spatial anomaly detection and clustering.
- **Data Visualization:** Conducted exploratory data analysis (EDA) and visualized complex spatial datasets in **QGIS** to validate model outputs and identify inconsistencies in land degradation metrics.
- **Reproducibility & Integrity:** Engineered a unified, documented data architecture for research backups, facilitating automated data retrieval and version control. This system ensured the consistency and completeness of large-scale datasets for multi-year research.

Projects

Deep Learning Foundations and Concepts Book Examples Reproduction

- Self-initiated project to reproduce key examples from C. Bishop and H. Bishop's *Deep Learning Foundations and Concepts* to gain theoretical and fundamental understanding of **polynomial regression** and **probability theory**.

aiSim Data Integration for 3D Gaussian Splatting

- Implemented a custom JSON parser to extract and sensor data obtained by aiSim, an autonomous driving simulator, into a nerfstudio format.

Skills

Programming: Python, Julia, C/C++, R, MATLAB
Tools: Docker, Linux, Bash, QGIS, Git, LaTeX